BUSINESS STUDY

Vehicle Rental Management System

Business problem • Process analysis • Requirements • Success measures

Prepared by	Shriya Verma
Project	Project 08 — Python OOP
Application	Menu-driven Python prototype using JSON

Purpose: This study explains the business need and proposed workflow for a small vehicle-rental operation. It is a project study, not a report of measured results from a real company.

1. Executive Summary

A vehicle-rental business needs a reliable way to track its fleet, know which vehicles are available, record customer rentals, calculate charges, and process returns. Manual notes or separate spreadsheets can lead to inconsistent status updates, calculation errors, duplicate records, and difficulty reviewing rental history.

The Vehicle Rental Management System is a menu-driven Python application that demonstrates a structured workflow for adding and viewing vehicles, renting and returning vehicles, calculating charges, and viewing rental records. JSON files are used to save and reload information between program runs.

Business value: The prototype demonstrates a more organised workflow and provides a foundation for future development. Its benefits have not been measured in a live business environment.

2. Business Problem

Potential problem	Possible business impact
Vehicle status is not updated consistently	Staff may offer a vehicle that is already rented.
Rental details are stored in multiple places	Records can be incomplete, duplicated, or hard to locate.
Charges are calculated manually	Calculation errors may cause customer disputes or lost revenue.
Returns are not clearly recorded	Staff may not know which vehicles are ready to rent again.
Rental history is difficult to review	Managers have limited visibility into vehicle use and rental activity.

3. Business Objectives

• Maintain a list of registered vehicles and their current availability.

• Record customer, vehicle, rental duration, calculated charge, and rental status.

• Update vehicle availability when a rental is returned.

• Preserve saved records in JSON files between program sessions.

• Provide a simple menu suitable for a user with basic computer skills.

4. Stakeholders and Their Needs

Stakeholder	Main need
Rental operator	Add vehicles, check availability, create rentals, and process returns.
Customer	Receive clear rental terms and an understandable rental charge.
Owner / manager	Review rental history and maintain accurate fleet status.
Developer	Keep the code understandable and extensible.

5. Proposed Business Process

1. Register vehicle: Enter a unique ID, model, daily rate, and seating capacity.

2. Check fleet: Review registered vehicles and their availability.

3. Start rental: Select an available vehicle, enter the customer's name and rental days, and calculate the charge.

4. Save transaction: Save the rental record and mark the vehicle as rented.

5. Process return: Mark the active rental as returned and make the vehicle available again.

6. Review history: View rental records and statuses.

6. Functional Requirements

ID	Requirement	Expected behaviour
FR-01	Add vehicle	Accept valid details and reject a duplicate vehicle ID.
FR-02	View vehicles	List vehicles, daily rates, and availability.
FR-03	Rent vehicle	Rent only a known vehicle marked available.
FR-04	Calculate charge	Multiply daily rate by rental days.
FR-05	Return vehicle	Update rental status and vehicle availability.
FR-06	View records	Display customer, vehicle, duration, charge, and status.
FR-07	Persist data	Load and save information using JSON files.
FR-08	Handle errors	Show helpful messages for invalid input and common file errors.

7. Business Rules and Assumptions

• Each vehicle must have a unique ID.

• A vehicle can be rented only when marked available.

• Rental duration must be a positive whole number of days.

• Daily rental rate must be greater than zero.

• The basic charge is daily rate × rental days. Taxes, deposits, discounts, insurance, and late fees are outside the current scope.

• This is a local, single-user prototype. It does not provide authentication, multi-user concurrency, online payments, or a booking website.

• JSON files are stored in the program's current working directory and should be backed up if records are important.

8. Suggested Success Measures (KPIs)

These are suggested measures for a future business deployment. No actual performance results are claimed.

KPI	Definition	Why it matters
Fleet availability	Available vehicles ÷ total vehicles × 100%	Shows the share of the fleet ready to rent.

KPI	Definition	Why it matters
Vehicle utilisation	Rental days ÷ available fleet-days × 100%	Indicates how effectively the fleet is used.
Rental revenue	Sum of rental charges for a period	Tracks rental income.
Average rental duration	Total rental days ÷ number of rentals	Helps understand rental patterns.
Return accuracy	Correctly updated returns ÷ total returns × 100%	Measures reliability of return processing.

9. Risks, Limitations, and Recommendations

Risk / limitation	Recommendation
Local JSON files can be lost or damaged	Back up files; consider a database as the project grows.
No login or permissions	Add authentication and role-based access before real business use.
No date-based availability checks	Store rental start/end dates and validate overlapping bookings.
No taxes, deposits, or late fees	Define business rules and test calculations before extending charges.
Vehicle-specific fields may not all persist yet	Test that seating capacity and all other fields are saved and restored.
Not designed for simultaneous users	Use a database with transaction and concurrency controls for production.

10. Conclusion

The Vehicle Rental Management System translates common rental operations into a structured menu-driven workflow. It demonstrates how an application can organise vehicle availability, rentals, returns, charge calculations, and history. It is suitable as a learning and portfolio prototype. Before real business use, it needs stronger validation, date-based booking logic, secure access, reliable storage, backups, and automated testing.

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Project: Project 08 — Vehicle Rental Management System
Tools: Python, Jupyter Notebook, JSON